

SEPTEMBER 2023 • SEQUOIA CAPITAL ANALYSIS

# **Generative AI's Act Two: From Technology Hammer to Real Value**

An analysis of the transition from novelty demos to end-to-end solutions solving real human problems.

# Six Decades of Technological Progress Converged to Enable Generative AI

Generative AI's emergence is the culmination of decades of compounded progress across three distinct waves:

- **Six Decades of Moore's Law**  
Delivered the necessary compute horsepower to process exaflops of data.
- **Four Decades of the Internet**  
Accelerated by COVID, producing trillions of tokens of training data.
- **Two Decades of Mobile and Cloud Computing**  
Put a supercomputer and distribution mechanism in every human's palm.

# GenAI Startups Surpassed \$1B Revenue Faster Than SaaS Ever Did

Despite hype-cycle skepticism, generative AI has proven unprecedented commercial demand, reaching the \$1 billion milestone in months rather than years.

**#1**

FASTEST GROWING APP  
ChatGPT found immediate product-market fit among developers and students.

**\$100M+**

MIDJOURNEY REVENUE  
Reportedly achieved with a lean team of just 11 people.

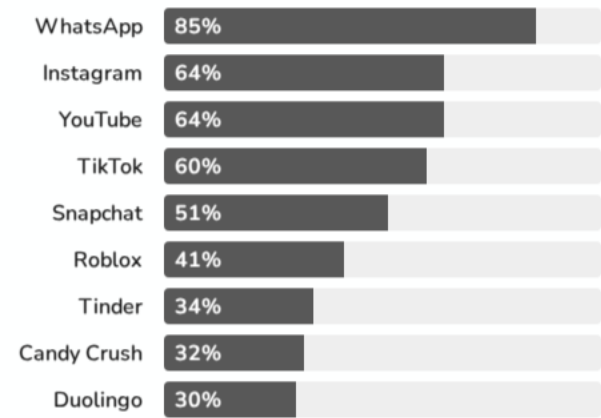
**2 Hrs**

AVG. SESSION TIME  
Character.ai popularized AI entertainment with massive user engagement.

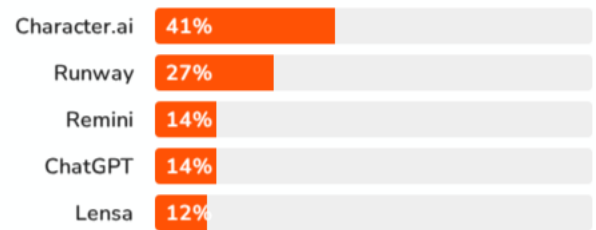
# AI-First Apps Show 14% Median Retention vs. 51% for Incumbents

Mobile app retention data reveals a stark 3.6x gap between established consumer apps and AI-first novelties.

## Incumbents (Median: 51%)

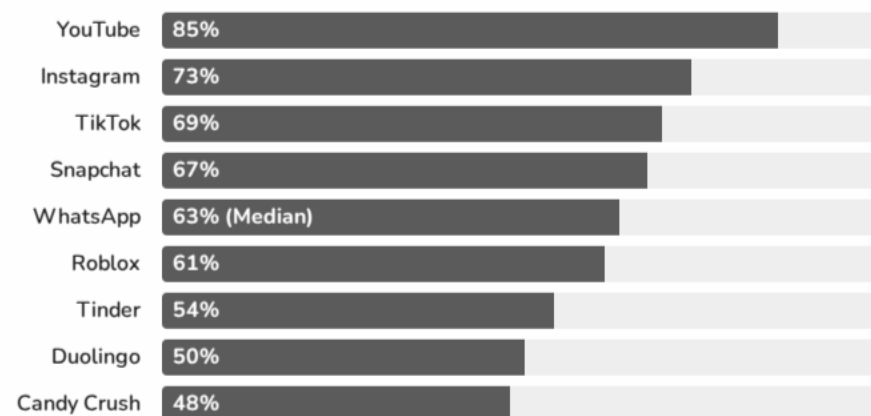


## AI-First Apps (Median: 14%)



# YouTube Leads Incumbents at 85% Monthly Active Usage

Monthly active usage benchmarks for incumbents show even higher engagement bars, contextualizing the immense challenge AI-first applications face.



# Act 1 Was Technology-Out; Act 2 Is Customer-Back

## Act 1: Technology-Out

- Foundation models as the **entire solution**.
- Novelty apps and lightweight demonstrations.
- Thin wrappers lacking stickiness.
- Characterized by poor user retention.

## Act 2: Customer-Back

- Models as **a piece** of a comprehensive solution.
- New editing interfaces for stickier workflows.
- Multi-modal and end-to-end problem solving.
- **Exemplars:** Harvey, Glean, Character, and Ava.

# Market Map Reorganized by Use Case, Not Model Modality

Reflecting the transition from technology hammer to actual value delivery, the generative AI market map is fundamentally shifting.

## From Modality to Value

The focus is moving away from the underlying models (text, image, code) and toward the actual end-user use case, reflecting the increasingly multi-modal nature of AI applications.

### Enterprise Examples

- **Harvey:** Custom LLMs for elite law firms
- **Glean:** Crawling & indexing enterprise workspaces

### Consumer Examples

- **Character & Ava:** Digital companionship and entertainment

## Key Takeaways: What Separates Winners in Act Two

- **Retention is the critical challenge:** AI-first apps suffer a severe 3.6x disadvantage (14% median) compared to incumbent consumer apps (51%).
- **Models are components, not products:** Foundation models must be a piece of the puzzle rather than the entire offering.
- **Multi-modal workflows win:** Integrating multiple capabilities into sticky, editing-focused interfaces creates defensibility.
- **Real demand exists:** Reaching the >\$1B revenue milestone so rapidly proves commercial viability beyond the hype cycle.
- **End-to-end solutions define Act 2:** Companies that work backward from human problems rather than forward from new technology will dominate the next era.